

Project GARUDA

Geospatial AI for Resilient Urban Disaster Analytics — गरुड, the all-seeing eagle that watches over and guards India's cities from above

A complete, research-grade architecture for ISRO BAH 2026 PS4. This document explains every component — **what, why, how, and the consequences** — and ties each one to a real problem and to what ISRO is doing today (NISAR, NDEM, NRSC). Prepared 25 Jun 2026 · v1.1 improvements 26 Jun 2026. Companion to the Deep Research Dossier.

0 · The reframe (the single most important decision)

PS4 literally asks for: occlusion-robust road extraction → graph healing → criticality (Gatekeeper Nodes) → stress testing → dashboard. **That is the baseline ~80% of teams will build.** We do not change the problem statement — we change its **altitude**.

From: "Detect roads from satellite images, even under trees/clouds." **To:** "Predict how a city's mobility network fails under a disaster — before it happens — and tell emergency responders exactly which junction, bridge or road to protect, and how to reroute when it breaks."

GARUDA turns multi-source satellite imagery (including ISRO's new **NISAR** radar) into an **uncertainty-aware, routable infrastructure graph**, fuses it with **structural-risk and hazard layers**, runs **failure simulations**, and delivers **NDEM-style decision support**. Every road we extract exists to answer one question: *"if this fails, who gets cut off, and what do we do?"*

WHY THIS FRAMING WINS (GROUNDED IN OUR RESEARCH)

The idea phase cut ~99.7% of teams in 2025 (8,744 teams → 30 finalists). Judges are **NRSC scientists whose day job is turning satellite data into disaster decision-support (NDEM)**. A road map impresses no one in that room; a **decision-support resilience platform built on their newest satellite (NISAR) and their operational paradigm (NDEM)** speaks their language. It is faithful to PS4, but operates at the level ISRO would actually deploy.

0★ · The one thing to remember (read this first)

The human story. In the **2015 Chennai floods**, 100+ patients — many on ventilators — were trapped inside Global Hospital, **cut off from the city for 48 hours** and finally evacuated by **fishing boats**; at MIOT Hospital,

14 patients died when isolation cut their power and oxygen. Responders could see where the water was. What no one could see in time was *which single road and bridge failures had severed those hospitals from the city*. **GARUDA exists to see that — before it becomes a tragedy.**

The one line

"GARUDA turns India's satellites into a living resilience intelligence system — predicting which single failure will cut a hospital off from the people who need it, and acting before the flood arrives."

★ **GARUDA's Universal Mission:** *"No Indian community should become digitally invisible — before, during, or after a disaster."*

Roads. Hospitals. Power. Water. Telecom. Schools. Relief. Every consequence flows from this one mission. GARUDA begins with urban road resilience and scales into **India's National Infrastructure Intelligence Platform**. The mission is not "detect roads" — it is **guaranteed digital visibility for every Indian community in crisis**.

★ The signature: the Zero-Isolation Engine

If a judge forgets everything else, they remember this. GARUDA's mission is **Zero Isolation** — *no hospital, shelter or community in India should ever be cut off by a single, predictable failure*. The **Zero-Isolation Engine** continuously finds every single-point-of-failure whose loss isolates a critical facility, and recommends the **minimum-cost reinforcement** that guarantees a backup route. Memorable, measurable, and the soul of the project.

The hero demo — one story, start to finish

Pre-monsoon rain builds → **NISAR InSAR** flags a weakening flyover embankment → GARUDA's living twin simulates the coming flood → the Zero-Isolation Engine detects this one failure **isolates City Hospital from ~150,000 residents** → **Guardian Mode** pushes an alert **36 hours early** → it recommends reinforcing one bridge and opening an evacuation corridor → officials act → the hospital stays connected. *That* is the demo.

City Pulse — the number people remember

GARUDA outputs one unforgettable figure per city — the **City Pulse**, a live Resilience Score (0–100) — and headline statements like *"this single flyover failure strands ~150k people and 1 hospital; reinforcing one bridge restores ~73% of cut-off access."*

Numbers are produced live by the simulation on the pilot city — illustrative here, never assumed.

★ Time-to-Isolation (TTI) — the decision metric no one else has

Alongside the City Pulse score, GARUDA introduces a new operational metric: **Time-to-Isolation (TTI)** — how many hours until a critical facility becomes unreachable if current infrastructure conditions continue deteriorating. Unlike a 0–100 score, TTI is immediately actionable. A score of 72/100 tells a planner "things are bad." **TTI = 9 hours tells them what to do in the next hour.**

Ward 8 · InSAR deformation trend + flood forecast →

□ City Hospital TTI = **9 hours** → Pre-position ambulances

NOW

□ Central School TTI = 17 hours → Alert and pre-evacuate by afternoon

□ Fire Station 3 TTI = 8 hours → Reroute access immediately

City Pulse score: 72/100 Critical Infrastructure: Hospital A
TTI 9h → Bridge Jn-14

GARUDA outputs both: the score for situational awareness, TTI for operational command. ISRO scientists haven't seen infrastructure risk presented this way — that's the point. **PoC** · TTI computed from InSAR deformation trend rate + flood inundation model + graph accessibility

Our research contribution (one sentence, scientist-clear)

We introduce a **confidence-aware resilience graph** that combines **satellite-derived infrastructure health** (NISAR InSAR temporal deformation), **uncertainty-aware road extraction** (optical-SAR fusion), and **network resilience simulation** (percolation + accessibility) into a single operational decision model — predicting and preventing critical-facility isolation before disaster strikes.

The national vision

GARUDA starts with roads in one city and scales into **India's National Infrastructure Resilience Platform**: roads → railways → power → water → telecom → health → the whole city → every Indian city. One nervous system for national resilience — the kind of indigenous, mission-grade capability ISRO builds.

Signature capabilities (useful features judges remember)

- **Zero-Isolation Engine** — guarantees no critical facility is one failure from being cut off; outputs the min-cost fix. **PoC**
- **City Pulse + TTI** — a live Resilience Score AND a Time-to-Isolation prediction per critical facility. **PoC**
- **Guardian Mode** — always-on proactive sentinel that warns before officials ask. **PoC**
- **"Next Failure" Panel** — Mission Control's signature screen: predicts the next infrastructure failure with confidence, TTI, population at risk, and recommended action — before anyone asks. **PoC**
- **Infrastructure Memory** — GARUDA doesn't just know what exists; it knows what has *changed*. NISAR's 12-day repeat cycle builds a temporal record: bridge deformation over 6 months, road subsidence over a year, flood recurrence over 5 years, urban expansion over 3 years. Only possible with ISRO's satellites. **Road** · **PoC**: *archived Sentinel-1/NISAR time-series for one bridge corridor*
- **Recovery Intelligence** — closes the full disaster lifecycle: post-event SAR damage assessment → repair priority ranking → recovery time estimation → reconstruction monitoring → before/after validation. **Road**
- **Pre-positioning advisor** — where to stage boats/ambulances given predicted isolations. **Road**
- **Pre-monsoon resilience audit** — automatic city vulnerability + priority-repair report before each monsoon. **PoC**

- **Investment Optimization** — "If you strengthen these 4 bridges, city resilience increases 18%." Answers the ₹50 crore question. PoC
- **Explainability** — "Why is this bridge critical?" → because 47% of hospital traffic passes through it; the nearest alternative adds 23 minutes; InSAR confidence 93%. Builds government trust. PoC
- **Confidence-aware routing** — emergency routes avoid low-confidence and high-risk segments. PoC

1 · Design principles (the rules every component obeys)

Principle	What it means in GARUDA	Why it matters for selection
Decision-first	Every layer ends in a decision a planner/responder can act on, not just a metric.	Judges score "does it solve the problem?" — not "is the IoU high?"
Uncertainty-honest	We output <i>confidence</i> , not just a road mask; we never silently hallucinate roads.	Safety-of-life decisions need calibrated trust; scientists distrust over-confident AI.
Sensor-honest	Each satellite is used for what its physics is actually good at (see §4).	An ISRO jury instantly spots teams who misuse SAR/optical. Correct usage = credibility.
Frugal & indigenous	Open + ISRO data; runs on Colab-grade GPU; "Atmanirbhar" data stack.	ISRO's real constraint is budget/cadence — frugal, India-data solutions score higher.
NDEM-aligned	Output mirrors ISRO's operational disaster DSS, ready to plug in.	Positions us as ISRO's next-gen tool, not a disconnected demo.
Explainable by design	Every risk rank, TTI score, and recommended action comes with a human-readable reason. Government trust requires explanation, not just prediction.	XAI for Earth Observation is a live research frontier (2024–25); emergency managers cite explainability as essential for operational adoption.
	Ambitious vision with an explicit PoC-vs-roadmap line (§9).	

Beyond-baseline, but buildable		Scientists reward credible scope; over-promising is punished.
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2 · The problems we solve (and exactly how)

PS4's stated pains, plus the deeper gaps ISRO faces — each mapped to the component that solves it.

Problem	Why it hurts	GARUDA's solution	Layer
Spectral blindness / occlusion (cloud, tree canopy, building shadow)	Optical sensors miss roads → broken, unusable masks (PS4's core pain)	Optical-SAR fusion: radar penetrates cloud/canopy to recover hidden roads	L1
AI hallucination / over-trust	A confident-but-wrong road can misroute an ambulance	Uncertainty quantification: per-pixel confidence + occlusion probability	L2
Fragmented masks ≠ routable network	Disconnected pixels can't compute shortest paths or detours	Confidence-weighted topological healing → routable weighted graph	L3
No structural/condition awareness	A road that exists may still be failing (subsidence, weak bridge)	NISAR InSAR deformation → per-edge structural vulnerability	L4
No temporal intelligence	Snapshots miss slow-developing failures — bridges deform over months before collapse	Infrastructure Memory: multi-pass InSAR time-series tracks change over months/years	L4+
No criticality / resilience analysis	Planners don't know which junction is a single point of failure	Betweenness + articulation points + TTI → Gatekeeper Nodes + Time-to-Isolation	L5
Reactive disaster response	NDEM monitors & assesses damage; less		L6

	<i>pre-emptive</i> failure prediction	Resilience digital twin + scenario engine: what-if BEFORE disaster	
No recovery intelligence	After the disaster, there's no satellite-native system to assess damage, rank repairs, or monitor reconstruction	Recovery Intelligence: SAR damage assessment → repair prioritization → reconstruction monitoring	L6+
Dashboards that show, not advise	A map of roads isn't an action for a control room	Decision dashboard + Next Failure panel: reroutes, TTI, priority-repair, isolated-facility alerts	L7
Data gating / cost / feasibility	Glamorous ideas die on unavailable data and budgets	Open + indigenous data, frugal compute, strict PoC scope	L0 + design

3 · The end-to-end architecture (master view)

Eight working layers from raw satellite data to a control-room decision, plus one roadmap layer.

purple = where ISRO's NISAR/SAR does the heavy lifting

L0 **Multi-source EO ingestion & harmonization** — optical (Cartosat-3, LISS-IV, Sentinel-2) + **SAR (Sentinel-1, NISAR)** + DEM + OSM + IMD rainfall. Co-register, tile, speckle-filter, normalize. **Multi-temporal stacking** for Infrastructure Memory baseline.



L1 **Occlusion-robust road extraction** — transformer segmentation (SegFormer/Mask2Former, ViT) with a **dual-stream optical+SAR fusion** encoder. Recovers roads under cloud/canopy/shadow.



L2 **Uncertainty & occlusion intelligence** — per-pixel road confidence + missing-segment/occlusion probability (evidential DL / MC-dropout / ensemble). *Certainty, not hallucination.*



L3

Topological healing → **routable graph** — skeletonize → nodes/edges; **MST + Disjoint-Set** gap-bridging weighted by confidence + geometric priors. Output: connected weighted graph (cost + confidence + structural risk).



L4 **Structural-risk fusion (NISAR InSAR) + Infrastructure Memory** — cm-level deformation → subsidence/bridge/embankment risk; SAR flood-water mapping; DEM landslide susceptibility → **per-edge/node vulnerability scores**. **Multi-pass time-series** builds Infrastructure Memory: what has changed over 6 months, 1 year, 5 years.



L5 **Criticality, accessibility & TTI engine** — betweenness centrality + articulation points → **Gatekeeper Nodes**; facility accessibility (hospitals/shelters via OSM) → "who is isolated if X fails." **TTI computation**: deformation trend rate + hazard model + graph accessibility → hours-to-isolation per critical facility.



L6 **Resilience digital twin, scenario engine & Recovery Intelligence** — overlay hazard → ablate nodes/edges → recompute **global network efficiency** → **Resilience Index**; emergency reroutes, evacuation corridors, priority-repair, lead-time alerts. **Investment Optimization**: greedy budget allocation for max resilience gain per ₹. **Recovery mode**: post-event SAR damage mapping → repair sequence → reconstruction tracking.



L7 **Decision-support dashboard (NDEM-aligned) + Mission Control** — criticality heatmap, TTI alerts, scenario toggles, **Next Failure panel**, recommended actions, impact report; export to NDEM/Bhuvan paradigm. **Explainability layer**: every ranked item has a plain-English "why."



L8 **(Roadmap, not core)** on-prem / air-gapped LLM planning copilot — natural-language "what-if" queries + auto incident reports. Shown as future vision.

TWO FLOWS IN ONE (THIS DISTINCTION WINS THE PROCESS-FLOW + ARCHITECTURE SLIDES)

Build-time pipeline (L0 → L3): heavy ML, run offline / pre-finale on archived data. **Run-time decision loop** (L4 → L7): lightweight graph + simulation, runs live in the control room on CPU. Separating these proves we know what's expensive vs what's interactive.

3★ · GARUDA is a closed-loop Resilience OS — not a linear pipeline

A sophisticated pipeline (data → AI → graph → dashboard) is still *passive and linear*. The leap scientist-judges reward is an **adaptive, self-improving operating system** for infrastructure resilience. Positioning matters: Windows

is an OS, not an app — **GARUDA is India's Infrastructure Resilience OS**. The city's road network is its first organ; the OS is built to host more (see §12★).

The OS loop: GARUDA runs a continuous **SENSE** → **UNDERSTAND** → **FORESEE** → **ACT** → **LEARN** cycle that feeds back on itself — it does not end at a dashboard.



The three feedback loops that make it adaptive (scientists love this)

1 · Active re-acquisition — the system decides what to look at next. When the scenario engine finds a *high-impact but low-confidence* failure point, it requests targeted re-evaluation of that area → uncertainty drops → graph & risk scores update → the simulation re-runs. GARUDA actively closes its own blind spots.

Roadmap · PoC: one manual re-evaluation trigger

2 · Living Digital Twin — continuous data assimilation. The twin is not a snapshot. It continuously ingests new satellite passes, **NISAR InSAR deformation**, IMD rainfall/forecast and traffic, and **auto-updates its state** (the same data-assimilation principle behind numerical weather models). Criticality, TTI, and the Resilience Index recompute as reality changes. **PoC*** · *lightweight: 1–2 feeds refresh the twin*

3 · Human-in-the-loop & learn-from-outcomes. Planners confirm/reject healed roads and actions → priors improve. After a real event, the twin checks whether its prediction held (TTI accuracy, isolation predicted vs. observed) and updates — so GARUDA gets **smarter with every disaster**. **Roadmap**

Passive dashboard → proactive intelligence

A passive system waits for a click. GARUDA's **Proactive Risk Sentinel** runs always-on, continuously scanning plausible hazards and **pushing the top risks before anyone asks**: "*City Hospital isolates if the Hebbal flyover floods — TTI = 11 hours — pre-position boats in Ward 7 now.*" That is intelligence, not visualization. **PoC**

GIS map → Mission Control (our product strength)

The UI is a **resilience mission-control room**, not a layer-toggle map: **Mission Status · Threat Level · Infrastructure Health Index · TTI Dashboard · Recommended Actions · Simulation Timeline · Resource Allocation**. It should feel like an ISRO control room. **PoC** core panels

★ "Next Failure" — the one panel no competitor has

Instead of showing current alerts, Mission Control's signature panel predicts *the next infrastructure failure before it occurs* — driven by InSAR deformation trend + rainfall forecast + TTI engine. This changes GARUDA from a monitoring tool to a **predictive command system**.

⚠ NEXT PREDICTED FAILURE

Infrastructure: Hebbal Overbridge

Confidence: **91%**

TTI: **18 hours**

Population at risk:

1,24,000

Hospitals cut off: **3**

Schools: 2

Fire Stations:

1

→ Recommendation: Pre-position rescue units in Sector 4 · Alert District Collector

PoC

· driven by InSAR deformation trend rate + IMD rainfall forecast + TTI engine

Live InSAR ingestion: Roadmap

The operational ecosystem — the architecture no longer ends at the dashboard

Mission Control → Mission Planning → Action & Resource rec. → Report Generation → Government API → Continuous Monitoring → ↻ Sense

- **Mission Planning** — pre-position resources, run drills, and *optimize where to invest* to raise the Resilience Index.

Road

- **Action & Resource recommendation** — ranked and **human-authorized** (never auto-dispatch).

Road

- **Report Generation** — one-click situation reports for officials.

PoC

- **Government API** — two-way integration with NDEM / Bhuvan / State DMAs.

Road

- **Continuous Monitoring** — always-on; closes the loop back to Sense.

Road

Discipline (we keep it — judges punish buzzword overload): the OS / ecosystem is the **vision**; the 30-hr **PoC stays tight** — extraction → living twin (1–2 feeds) → 1–2 scenarios → one proactive alert → mission-control dashboard with TTI panel and report export. Everything else is clearly tagged roadmap. We deliberately do **not** stack RL, blockchain, IoT, drones or generic "agents" — *every capability must earn its place*.

4 · Sensor-fusion rationale — using each satellite for what its physics is good at □

This section is what separates a credible ISRO submission from a buzzword deck. We do **not** claim NISAR "sees roads" (its resolution is ~3–10 m, tuned for deformation, not fine imaging). We use each sensor for its real strength:

Sensor	Physical strength	Role in GARUDA
Cartosat-3 (≤ 0.25 m PAN)	Sub-metre optical detail	Fine road geometry / centerlines (finale-supplied tile)
Resourcesat LISS-IV (5.8 m), Sentinel-2 (10 m)	Open, multispectral, frequent	City-scale road extraction + training data
Sentinel-1 C-SAR (10 m), NISAR L+S-SAR	All-weather; penetrates cloud & vegetation canopy; sensitive to structure & water	Occlusion recovery (roads under cloud/canopy) + all-weather flood/inundation mapping
NISAR InSAR (cm deformation, 12-day repeat)	Centimetre surface-motion detection over time — NISAR detects changes as small as 4 mm/year	Structural risk : subsidence, bridge/embankment deformation, landslide precursors → edge vulnerability. Infrastructure Memory : multi-pass time-series tracks slow-developing failures over months to years.
CartoDEM / SRTM	Elevation, slope, flow	Flood & landslide susceptibility modelling
OpenStreetMap	Crowd-sourced vectors + POIs	Auto ground-truth for training + hospitals/shelters for accessibility
IMD rainfall / forecast	Hazard trigger	Drives the flood scenario engine + TTI computation

The credibility line for judges: "Optical gives us the geometry; **SAR gives us the truth under clouds**; **NISAR's InSAR gives us the road's health over time** — we can see a bridge weakening over six months before it fails. We fuse what each Indian and open sensor does best." A scientist nods at that.

5 · Component deep dive — what · why · how · consequences

For each layer: exactly what it does, why it must exist, how we build it, what it unlocks (and what breaks without it), and how it ties to ISRO's real work.

LAYER 0

Multi-source EO ingestion & harmonization

What Pull and align all inputs into analysis-ready tiles: optical (Cartosat-3, Resourcesat LISS-IV, Sentinel-2), SAR (Sentinel-1, NISAR), DEM (CartoDEM/SRTM), OSM vectors+POIs, IMD rainfall. Also stack multi-temporal SAR acquisitions for Infrastructure Memory baseline.

Why Roads must be observable in every condition; a single sensor is blind under cloud/canopy. Every downstream layer assumes pixel-aligned, calibrated data — get this wrong and fusion produces noise.

How Sources: Bhoonidhi (ISRO), Copernicus, NOAA/AWS. Optical–SAR **co-registration** (ESA SNAP / GDAL / OTB); SAR **speckle filtering** (Refined Lee), radiometric + terrain correction; tile to $512^2/1024^2$; normalize. Pretraining pairs from **SEN1-2 / OpenSatMap**. Multi-temporal SAR stack: SBAS-InSAR or PS-InSAR processing for deformation time-series.

Consequences The silent foundation — good harmonization makes fusion trivial; bad alignment poisons everything. *Tradeoff*: SAR–optical co-registration is fiddly → we lean on pre-aligned public datasets for the PoC.

ISRO tie-in Runs on **Bhoonidhi** (44-satellite archive) + the **NISAR** portal; indigenous Cartosat/Resourcesat → an Atmanirbhar, India-data stack.

Occlusion-robust road extraction (optical–SAR fusion)

What A deep segmentation model that outputs road probability from **fused optical + SAR** input.

Why This is PS4's central pain — "spectral blindness." Trees, shadows and cloud create topological breaks; single-sensor extractors provably fail under occlusion. Radar penetrates cloud and canopy, so SAR supplies structure exactly where optical dies.

How Transformer segmentation (**SegFormer / Mask2Former**, ViT backbone) with a **dual-stream encoder** (optical + SAR) fused via cross-attention / discrepancy-conditioned fusion. Loss = Dice + IoU + **boundary-aware** + optional **connectivity loss**; train with **simulated occlusions**; pretrain on **SpaceNet + DeepGlobe**, fine-tune on the pilot city.

Consequences Big jump in **occlusion-recall** (the metric PS4 stresses) → continuous roads where baseline teams get gaps. *Tradeoff*: two streams cost more compute/data → bounded by PoC scope and pretrained backbones.

ISRO tie-in ISRO already runs AI-on-EO (deforestation reporting 1 yr → 1 mo; Cartosat-3 water-body DNN at 96%). SAR–optical fusion is peer-reviewed (GLF-CR, D3FNet). Uses NISAR's all-weather strength as designed.

Uncertainty & occlusion intelligence

What Per-pixel **road confidence** + **occlusion / missing-segment probability** alongside the mask.

Why A confident-but-wrong road can send an ambulance into a dead end. Planners must see *where the model is guessing*. Uncertainty also drives smarter healing (L3) and an honest UI (L7).

How **Evidential deep learning** (Dirichlet) or **MC-dropout / deep ensembles** → confidence map; **temperature-scaling** calibration; measure with **Expected Calibration Error**. Low-confidence / occluded segments are flagged distinctly, never silently filled.

Consequences Enables confidence-weighted healing and trustworthy decisions; calibration is quantifiable. Without it the platform *silently lies* — disqualifying for safety-of-life use.

ISRO tie-in Mirrors ISRO's heightened **reliability/quality-assurance** focus after the EOS-09 (18 May 2025) failure — responsible AI for governance.

Topological healing → routable graph

What Convert the pixel mask into a **connected, weighted, routable vector graph**.

Why PS4 is explicit: fragmented masks can't compute shortest paths, detours or failures. Topology is the bridge from "image" to "network."

How Morphological **skeletonization** → 1-px centerlines → nodes (intersections/endpoints) + edges. **MST + Disjoint-Set** bridge gaps using Euclidean distance + **angular alignment** + L2 confidence. Edge weight = travel cost (length/speed-limit); edge attribute = structural confidence.

Consequences Produces the object every downstream layer needs; PS4's **connectivity-ratio** metric jumps after healing. Confidence-gating stops us "healing" fake roads.

ISRO tie-in Implements PS4's prescribed MST/Disjoint-Set; yields an NNRMS-grade routable road asset.

Structural-risk fusion (NISAR InSAR) & Infrastructure Memory

What Enrich every edge/node with a **structural-vulnerability** score and **hazard-exposure** score. Build an **Infrastructure Memory** from multi-pass InSAR time-series — what has changed over months and years.

Why A road that *exists* is not a road that's *safe*. Bridges deform measurably for months before they fail; floods and landslides disable specific segments. **ISRO's greatest unique strength is repeated Earth observation** — no ground-based system can match NISAR's 12-day repeat, 4 mm/year sensitivity. A static snapshot wastes this advantage; Infrastructure Memory exploits it.

How **NISAR / Sentinel-1 InSAR** time-series (SBAS/PS-InSAR) → cm-level **deformation trends** (bridge/embankment/road subsidence motion); **SAR all-weather flood mapping**; **DEM** slope + flow → landslide susceptibility. Infrastructure Memory: build deformation velocity maps + anomaly detection (is this bridge sinking faster than it was 6 months ago?). Peer-reviewed 2025 research confirms InSAR clustering algorithms detect bridge displacement trends at mm accuracy.

Consequences Converts the graph into a **condition-aware, time-aware twin** → failure scenarios become evidence-based predictions, not random stress tests. The TTI engine pulls directly from Infrastructure Memory (deformation velocity → time-to-failure estimate). *Tradeoff*: InSAR time-series is heavy → PoC uses archived/derived data; live ingestion is roadmap.

ISRO tie-in NISAR's headline applications are **deformation + infrastructure + disaster**; echoes RISAT flood mapping (Wayanad, Jul 2024) and NDEM's **Hazard–Vulnerability–Risk** workflow. NISAR's 12-day repeat is uniquely suited to temporal infrastructure monitoring that no private or government data stream can replicate — the satellite-first advantage.

Criticality, accessibility & TTI engine

What Rank nodes/edges by systemic importance, compute facility access, and calculate **Time-to-Isolation (TTI)** per critical facility.

Why Planners need the **single points of failure**, the answer to "who loses hospital/shelter access if this breaks?" — AND how long they have to act.

How **Betweenness centrality** → **Gatekeeper Nodes**; **articulation-point** / cut-vertex detection; shortest-path **accessibility** from population grids to hospitals/shelters. **TTI engine**: for each critical facility, compute: (current edge vulnerability + deformation velocity from L4) / (accessibility loss rate under projected hazard) → hours until isolation. Combines Infrastructure Memory + hazard forecast + graph dynamics into one actionable number.

Consequences Produces the criticality heatmap + facility-isolation + TTI dashboard (PS4 + beyond). TTI is a new operational metric ISRO scientists haven't seen presented this way — that's the differentiation. Targeted (centrality-based) ranking is provably more disruptive than random → it focuses scarce attention where it matters.

ISRO tie-in Directly delivers PS4's "Gatekeeper Nodes"; TTI supports NDEM situational assessment and AMRUT urban-infrastructure planning. OECD/ITF 2025 transport resilience policy cites rapidity (time to failure/recovery) as a core resilience dimension — TTI operationalizes this.

Resilience digital twin, scenario engine & Investment Optimization

What Simulate disasters/failures on the twin, quantify resilience, generate actions, and **optimize infrastructure investments**.

Why PS4's stress-testing — and the **pre-emptive** capability ISRO's NDEM is weakest on. "Predict how the city fails *before* it fails." Also: government asks "we have ₹50 crore — where do we spend it?" GARUDA should answer that.

How Hazard overlay → disable affected nodes/edges (**percolation framework**) → recompute **global efficiency**, largest connected component, facility accessibility → **Resilience Index = perturbed ÷ baseline efficiency**. Generate: emergency reroutes, evacuation corridors, **priority-repair ranking**. **Investment Optimization**: greedy/combinatorial search — "if we strengthen these 4 bridges, overall city resilience increases 18%." Budget-constrained network robustness optimization (peer-reviewed: resilience-based component importance metrics + budget allocation, 2024).

Consequences The interactive, unforgettable demo: click "flood" → city reorganizes, hospital lights up as isolated, reroute appears. Investment Optimization turns GARUDA from a diagnostic tool into a **capital allocation advisor** for NDMA/municipalities — a genuinely different value proposition. *Tradeoff*: full combinatorial optimization is NP-hard → use greedy top-k bridge reinforcement for PoC.

ISRO tie-in Extends NDEM's HVR + decision-support with the pre-emptive layer it lacks; disaster resilience is the host center's (NRSC) core mandate. Investment optimization is precisely what AMRUT cities (500 cities, GIS master plans) need — a satellite-derived evidence base for infrastructure spend.

Decision-support dashboard (NDEM-aligned) + Explainability

What The control-room interface that delivers decisions, not just maps — with a full **Explainability layer** so every ranked item comes with a reason.

Why A model in a notebook helps no responder. Judges explicitly score "professional, clear presentation." Government officials will not act on a black-box ranking — they need to understand *why* before they authorize action. XAI research (2024–25) confirms that feature attribution and input-output visualization are essential for trust in emergency management AI.

How Web app (**Streamlit/React + Leaflet/Mapbox**): toggleable layers (road-confidence, criticality heatmap, hazard overlay), **TTI panel**, **Next Failure panel**, **scenario toggle**, **recommended-actions panel**, **impact report**, export (GeoJSON/PDF) into the NDEM/Bhuvan paradigm. **Explainability**: click any node/edge → see: "Why is this bridge critical? → 47% of hospital traffic passes through it. The nearest alternative adds 23 minutes. InSAR deformation confidence: 93%." Every TTI score, every ranked action, every investment recommendation comes with a plain-English explanation.

Consequences Makes GARUDA feel like deployable government software, not an experiment. Explainability builds the trust that moves officials from "interesting demo" to "we want to use this." The layer your team is strongest at and where decks win or lose.

ISRO tie-in Mirrors **Bhuvan NextGen "Disaster mode"** and NDEM DSS → framed as their next-generation tool. Prototype-based Explainable Neural Networks for geospatial learning tasks (2026 research) confirms XAI is the frontier for EO decision support.

On-prem / air-gapped LLM planning copilot

What Natural-language "what-if" queries + auto-generated incident reports over the twin.

Why Lowers the barrier for non-technical officials — but it is **not** the core claim. We flag it as future to keep the submission credible.

How Small **quantized on-prem LLM** + RAG over the twin's live state, runbooks and past incidents; air-gapped for government security.

Consequences Future differentiation; deliberately excluded from the 30-hr PoC so we never over-promise.

ISRO tie-in Matches government **air-gapped/offline** needs — responsible, sovereign AI.

5★ · Full feature catalog — every capability that earns its place

The judging template has a dedicated "Features" slide. Most teams list 5–6. We offer a deep, layered catalog — and the **small features** (🔪) **often win the room** because they prove we thought about the real user in a real disaster.

PoC = in the finale build · **Road** = roadmap.

A · Perception (extraction)

- Optical–SAR fusion road extraction (occlusion-robust) **PoC**
- Multi-temporal compositing to beat transient cloud **PoC**
- Per-pixel road confidence + occlusion-probability map **PoC**
- Missing-segment recovery (graph-level inpainting) **PoC**
- ⚡ Paved-vs-unpaved hint from SAR backscatter (flood usability) **Road**
- Bridge / overpass / culvert tagging · lane-count proxy **Road**

B · Network (graph)

- Confidence-weighted topological healing (MST + Disjoint-Set) **PoC**
- Routable weighted graph export (GeoJSON / OSM-compatible) **PoC**
- ⚡ Auto OSM change-detection → map-update suggestions **PoC**
- Bridge & articulation-point detection **PoC**
- ⚡ Per-link redundancy / alternate-route score **PoC**

C · Risk, Hazard & Infrastructure Memory

- NISAR InSAR deformation → per-edge structural vulnerability **PoC***
- SAR all-weather flood / inundation mapping **PoC**
- DEM flood-depth + landslide susceptibility **PoC**
- Historical-event overlay (validation + learning) **PoC**
- < **Infrastructure Memory**: multi-pass InSAR time-series → bridge deformation trends (6 mo), road subsidence (1 yr), flood recurrence (5 yr). What changed — uniquely satellite.
Road · *PoC: one bridge corridor*
- < Rainfall-triggered dynamic hazard (IMD live) **Road**
- < Waterlogging / bridge-scour hotspot flags **Road**

D · Criticality, Resilience & Investment Optimization

- Betweenness "Gatekeeper Node" ranking **PoC**
- Global-efficiency Resilience Index **PoC**
- **Time-to-Isolation (TTI)**: hours until each critical facility becomes unreachable **PoC**
- Critical-facility (hospital/shelter/fire/fuel) isolation analysis **PoC**
- < Population-at-risk per scenario (census / WorldPop grid) **PoC**
- Evacuation-corridor identification **PoC**
- < **Investment Optimization**: "If you strengthen these 4 bridges, city resilience increases 18%." Budget-constrained greedy network robustness optimization — answers the ₹50 crore question for NDMA/ municipalities. Turns GARUDA into a *capital allocation advisor*. **PoC** · *greedy top-k on the twin*
- Cascading-failure simulation (closure → congestion → secondary) **Road**

E · Foresight, Scenarios & Recovery Intelligence

- Interactive what-if: flood / landslide / quake / closure / festival **PoC**
- Scenario A/B comparison + before/after **PoC**
- < **"Next Failure" panel**: predicted failure + confidence + TTI + population at risk + recommendation **PoC**
- < Lead-time alert from InSAR deformation trend (days-to-risk) **Road**
- Multi-hazard compound scenarios · monsoon/ time-of-day modes **Road**
- < **Recovery Intelligence**: post-disaster SAR damage assessment → repair priority ranking → recovery time estimation → reconstruction monitoring → before/after comparison. Closes the full disaster lifecycle. **Road**

F · Decision, UX & Explainability

- Criticality heatmap + hazard overlays **PoC**
- TTI dashboard — hours-to-isolation per critical facility **PoC**
- Ranked recommended-actions panel · one-click reroute + ETA **PoC**
- Impact report export (PDF/GeoJSON) · < NDEM/Bhuvan-compatible **PoC**
- Confidence toggle ("show where AI is unsure") **PoC**
- < **Explainability**: click any node → "Why is this bridge critical?" → "47% of hospital traffic passes through it. The nearest alternative adds 23 minutes. InSAR deformation confidence: 93%." Every TTI, every recommended action, every investment ranking comes with a human-readable reason. Government trust requires explanation. **PoC**
- < Multilingual UI (Hindi + regional) — official accessibility **Road**
- < Offline / low-bandwidth field mode — crucial in disaster zones **Road**
- < Role-based views · alert hooks (SMS/ WhatsApp) · REST API **Road**

G · Trust & responsible AI

- Uncertainty calibration display (ECE) **PoC**
- ⚡ Human-in-the-loop verify/reject healed roads **PoC**
- Data provenance & timestamp per layer **PoC**
- No-hallucination policy (flag, never silently fill) **PoC**
- Explainable rankings — every prediction has a traceable reason **PoC**

H · Scale & sustainability (ISRO-fit)

- Indigenous-data-first (Bhoonidhi / NISAR / Cartosat) **PoC**
- ⚡ Self-supervised pretraining for new cities (low-label) **Road**
- Tiled multi-city → national scaling **Road**
- Frugal compute (Colab/Kaggle) · edge cloud-mask on-board **Road**
- Continuous learning from new satellite passes **Road**

*L4 InSAR runs in PoC on archived/derived deformation; live ingestion is roadmap. ⚡ = small feature, outsized impact — these are the details that signal real-world thinking to an NRSC jury.

6 · Data & datasets (all PoC data is open today)

Purpose	Source	Readiness
Pretraining (roads)	SpaceNet Roads, DeepGlobe, OpenSatMap	Open
Ground truth + POIs	OpenStreetMap (auto road labels; hospitals/shelters)	Open
Optical (geometry)	Sentinel-2 (Copernicus/GEE), Resourcesat LISS-IV (Bhoonidhi); Cartosat-3 at finale	Open / finale
SAR (occlusion + flood + Infrastructure Memory)	Sentinel-1 (Copernicus); NISAR (Bhoonidhi NISAR portal) — 12-day repeat InSAR time-series for deformation monitoring	Open (archived for PoC)
Elevation / hazard	CartoDEM / SRTM; IMD rainfall	Open

Pilot city: pick one cloud/flood-prone Indian metro with a documented historical disruption for validation (e.g., Bengaluru — PS4's named example; or Chennai-2015 / Guwahati for flood drama). One city, done deeply, beats five done shallowly.

7 · Technology stack (only what we will actually use)

Block	Tools
Geospatial	GDAL, Rasterio, ESA SNAP (SAR + InSAR), GeoPandas, OSMnx, scikit-image
Deep learning	PyTorch; SegFormer / Mask2Former (mmsegmentation/HF); Albumentations; MC-dropout / evidential layers
Graph & simulation	NetworkX (betweenness, efficiency, percolation, TTI engine, investment optimization); PyG/GNN (roadmap)
InSAR / time-series	ESA SNAP (SBAS-InSAR); StaMPS / MintPy (PS-InSAR for Infrastructure Memory)
Backend / twin	Python, FastAPI; PostGIS (optional)
Dashboard	Streamlit (fast PoC) or React + Leaflet/Mapbox — with explainability tooltips and TTI panels
Compute & data	Google Colab / Kaggle (T4/A100, free); Bhoonidhi, Copernicus, NOAA AWS, OSM, IMD

8 · Evaluation & metrics

PS4-PRESCRIBED (WE HIT EVERY ONE)

- **IoU & Dice** with explicit **occlusion-recall** (road recovery under shadow/canopy/cloud).
- **Generalisation** across dense-urban, suburban-forested and rural terrains.
- **Connectivity ratio** — % increase in the largest connected component after MST healing.
- **Topological accuracy vs OSM** — average shortest-path-length error between random point p